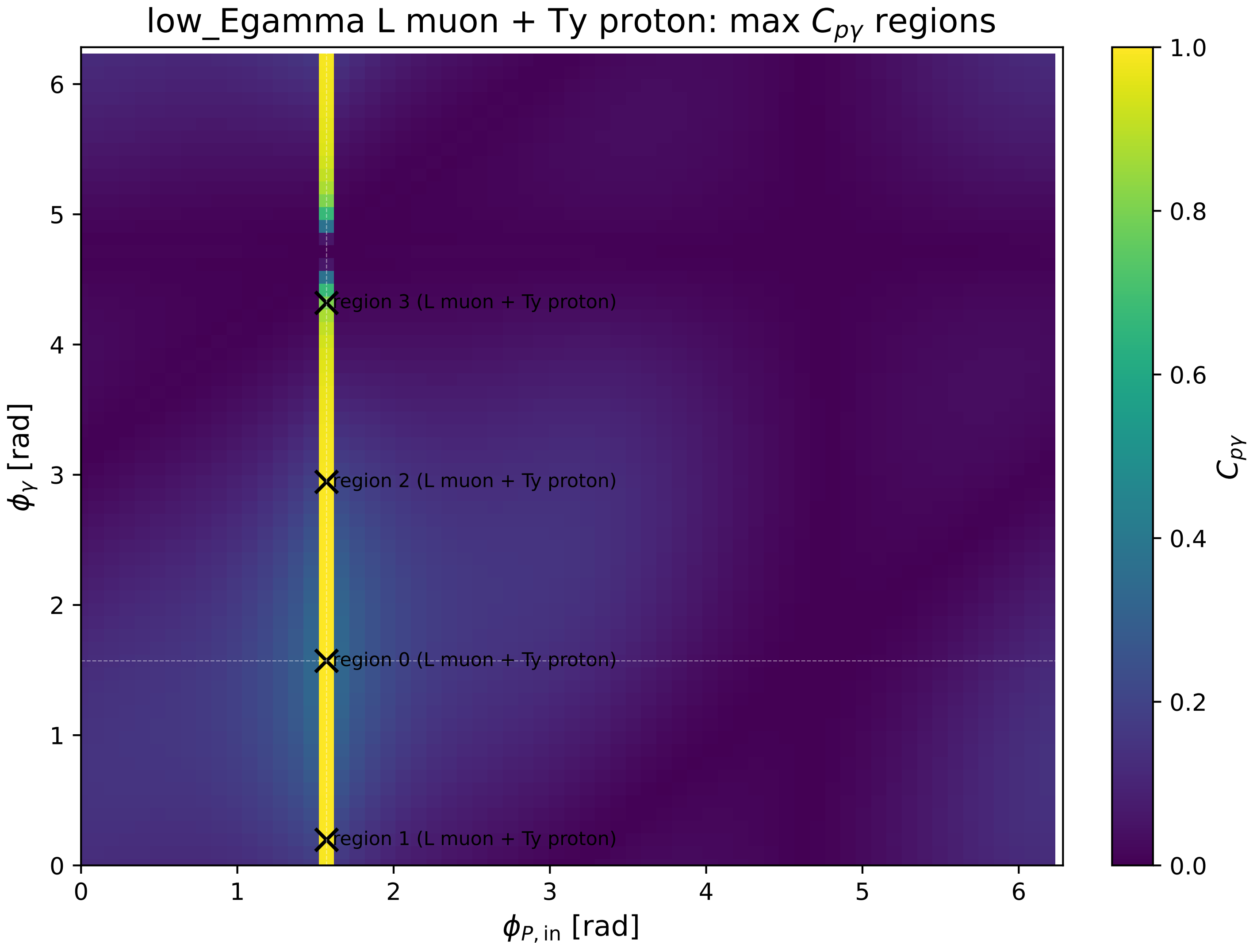
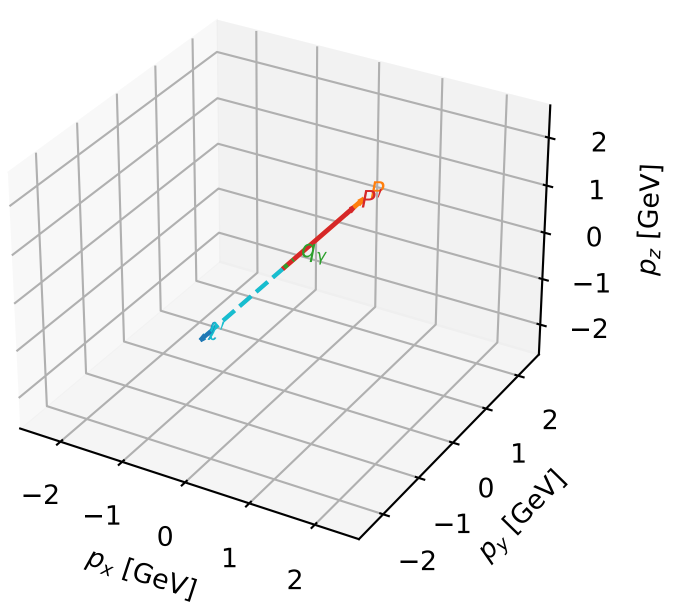


low_Egamma L muon + Ty proton: max $C_{p\gamma}$ regions
region 3 (L muon + Ty proton)
region 2 (L muon + Ty proton)
region 0 (L muon + Ty proton)
region 1 (L muon + Ty proton)
ϕ_γ [rad]
$\phi_{P,\mathrm{in}}$ [rad]
$C_{p\gamma}$
0
1
2
3
4
5
6
0.0
0.2
0.4
0.6
0.8
1.0

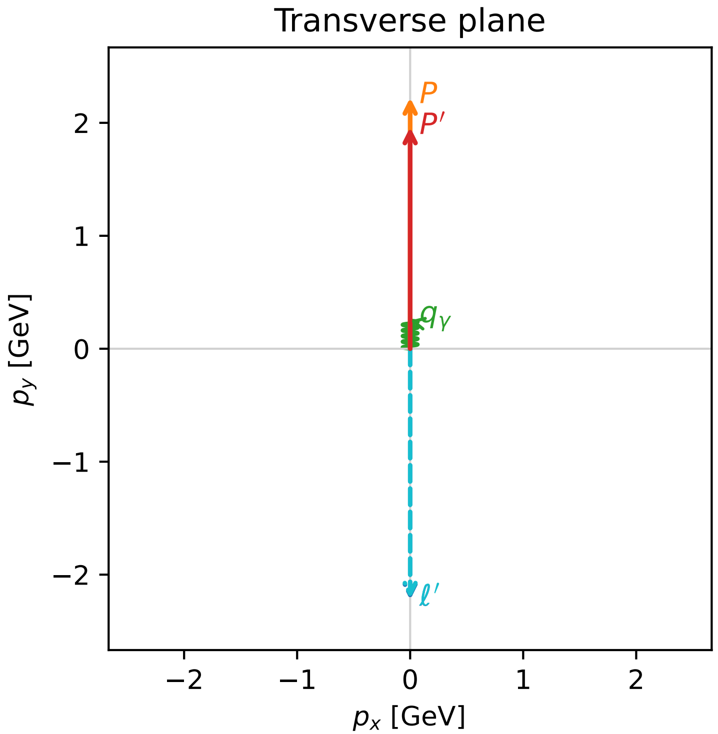
L muon + Ty proton characteristic max $C_{p\gamma}$ configuration

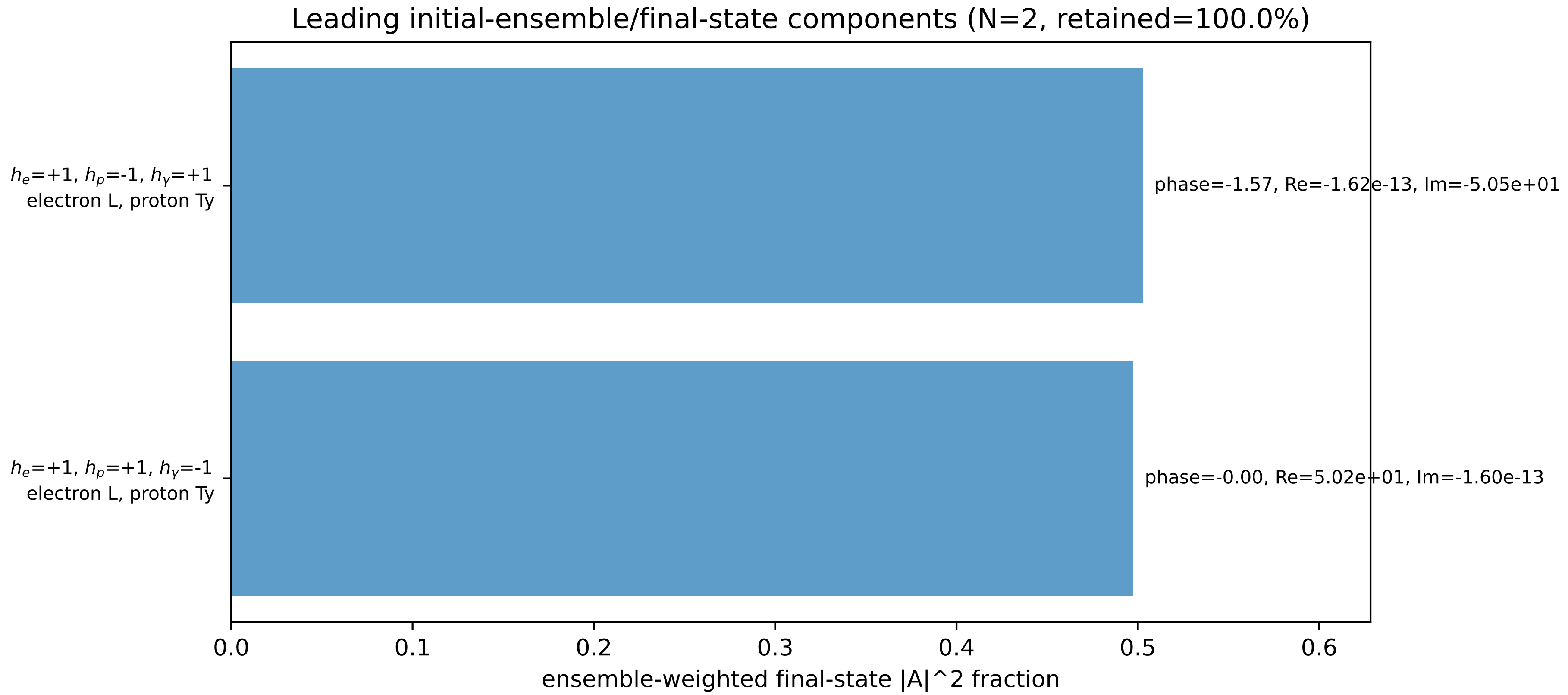


c_p_gamma_low_egamma_lty_region_0 C_{p\gamma}=0.999986
region: low_Egamma_LTy_region_0 final-pair delta_xy=0 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

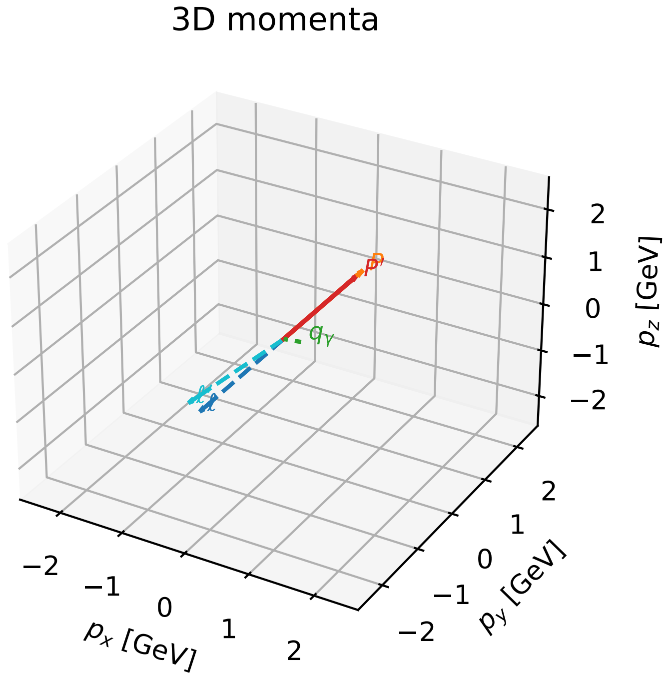
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=1.96163
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=1.5708
Q^2=2.98483e-07, x_B=2.80622e-06, t=-0.0114721, W^2=0.986208, y=0.00515711
$\theta(\ell', \gamma)$=180 deg

four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 2.2142 p_x= -0.0000 p_y= -2.2116 p_z= 0.0000
P' E= 2.1744 p_x= 0.0000 p_y= 1.9616 p_z= 0.0000
q_γ E= 0.2500 p_x= 0.0000 p_y= 0.2500 p_z= 0.0000





L muon + Ty proton characteristic max $C_{p\gamma}$ configuration

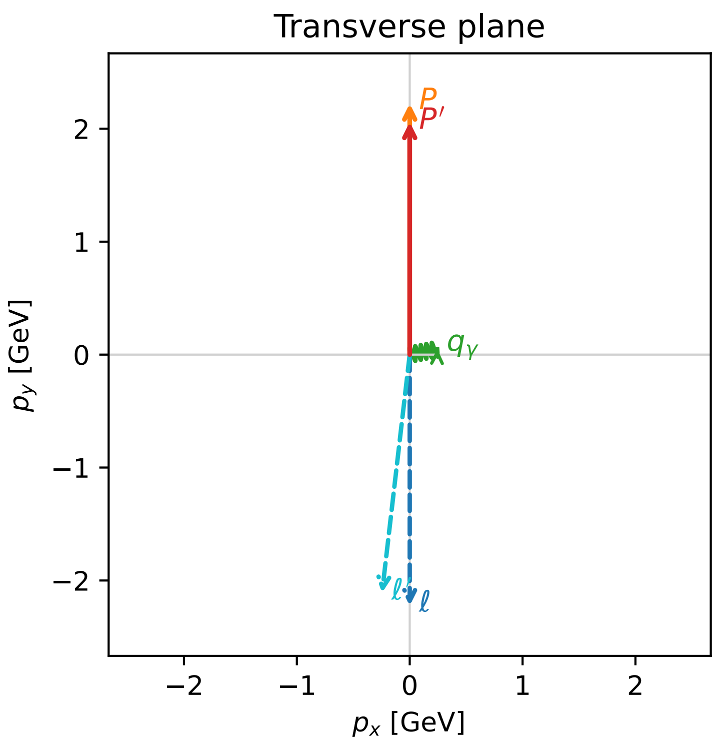


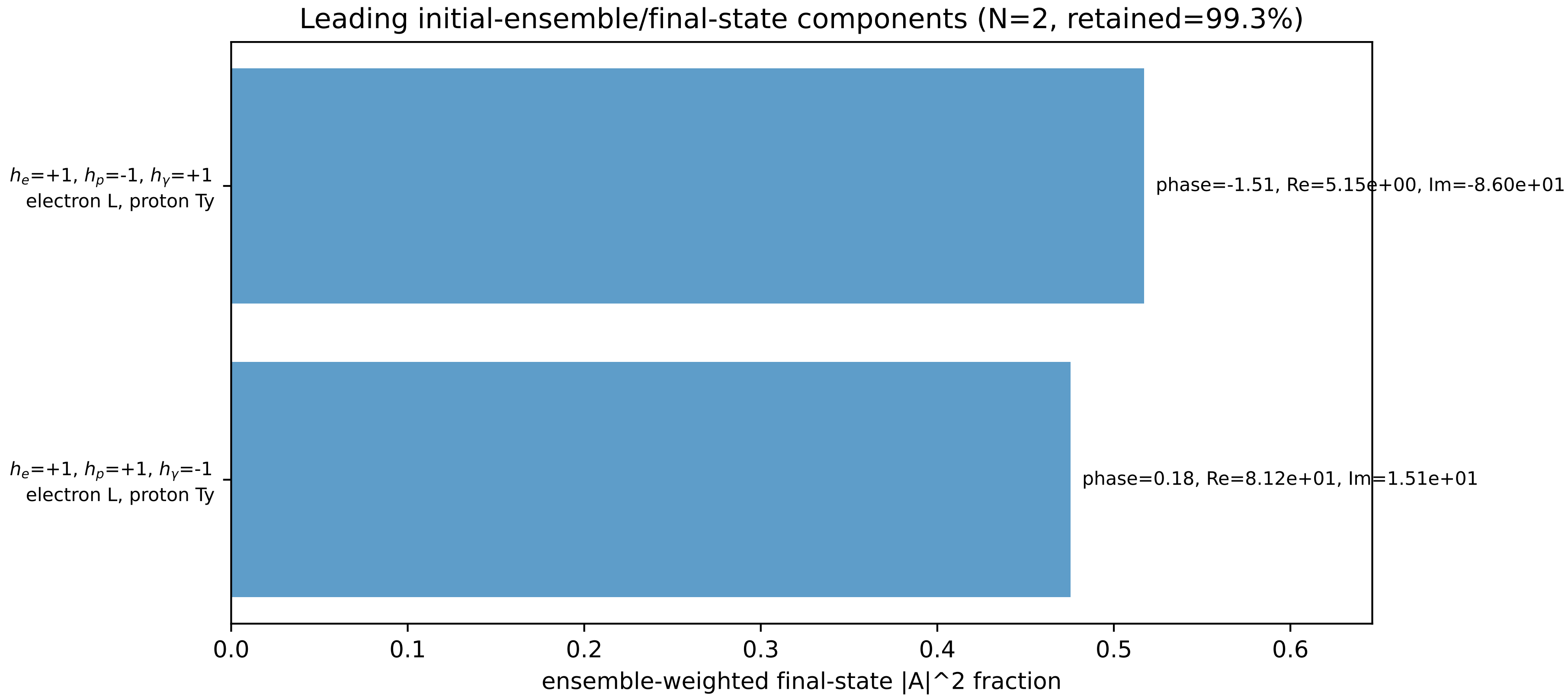
c_p_gamma_low_egamma_lty_region_1 C_{p\gamma}=0.991891
region: low_Egamma_LTy_region_1 final-pair delta_xy=1.37445 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$

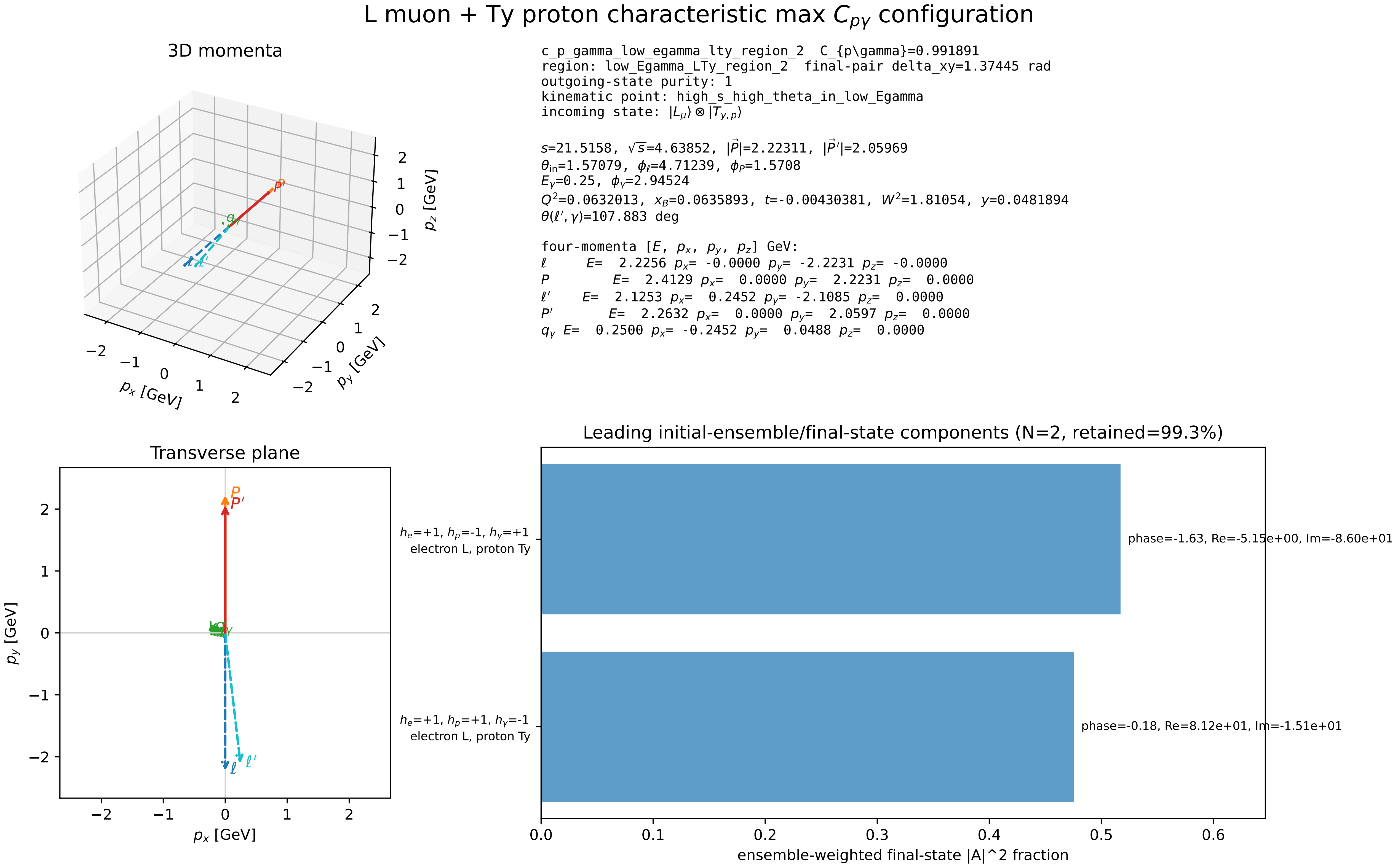
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.05969
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=0.19635
Q^2=0.0632013, x_B=0.0635893, t=-0.00430381, W^2=1.81054, y=0.0481894
$\theta(\ell', \gamma)$=107.883 deg

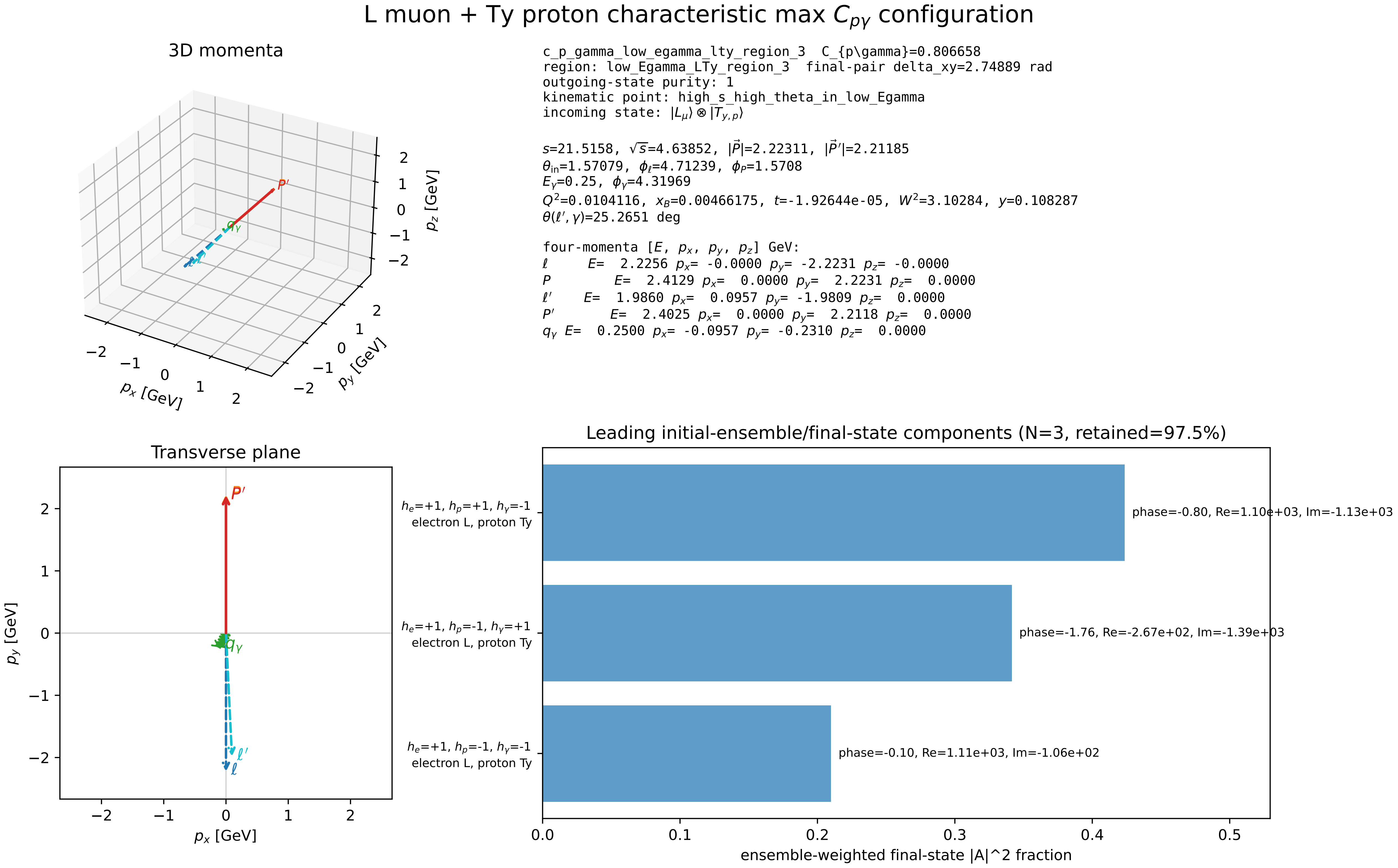
four-momenta [E, p_x, p_y, p_z] GeV:

ℓ	E= 2.2256	p_x= -0.0000	p_y= -2.2231	p_z= -0.0000
P	E= 2.4129	p_x= 0.0000	p_y= 2.2231	p_z= 0.0000
ℓ'	E= 2.1253	p_x= -0.2452	p_y= -2.1085	p_z= 0.0000
P'	E= 2.2632	p_x= 0.0000	p_y= 2.0597	p_z= 0.0000
q_γ	E= 0.2500	p_x= 0.2452	p_y= 0.0488	p_z= 0.0000






L muon + Ty proton characteristic max $C_{p\gamma}$ configuration
3D momenta
p_x [GeV]
p_y [GeV]
p_z [GeV]
c_p_gamma_low_egamma_lty_region_2 C_{p\gamma}=0.991891
region: low_Egamma_LTy_region_2 final-pair delta_xy=1.37445 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.05969
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=2.94524
Q^2=0.0632013, x_B=0.0635893, t=-0.00430381, W^2=1.81054, y=0.0481894
$\theta(\ell', \gamma)$=107.883 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 2.1253 p_x= 0.2452 p_y= -2.1085 p_z= 0.0000
P' E= 2.2632 p_x= 0.0000 p_y= 2.0597 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.2452 p_y= 0.0488 p_z= 0.0000
Transverse plane
p_x [GeV]
p_y [GeV]
Leading initial-ensemble/final-state components (N=2, retained=99.3%)
h_e=+1, h_p=-1, h_γ=+1
electron L, proton Ty
phase=-1.63, Re=-5.15e+00, Im=-8.60e+01
h_e=+1, h_p=+1, h_γ=-1
electron L, proton Ty
phase=-0.18, Re=8.12e+01, Im=-1.51e+01
ensemble-weighted final-state |A|^2 fraction



L muon + Ty proton characteristic max $C_{p\gamma}$ configuration
3D momenta
c_p_gamma_low_egamma_lty_region_3 C_{p\gamma}=0.806658
region: low_Egamma_LTy_region_3 final-pair delta_xy=2.74889 rad
outgoing-state purity: 1
kinematic point: high_s_high_theta_in_low_Egamma
incoming state: $|L_\mu\rangle \otimes |T_{y,p}\rangle$
s=21.5158, $\sqrt{s}$=4.63852, $|\vec{P}|$=2.22311, $|\vec{P}'|$=2.21185
θ_{in}=1.57079, ϕ_ℓ=4.71239, ϕ_P=1.5708
E_γ=0.25, ϕ_γ=4.31969
Q^2=0.0104116, x_B=0.00466175, t=-1.92644e-05, W^2=3.10284, y=0.108287
$\theta(\ell', \gamma)$=25.2651 deg
four-momenta [E, p_x, p_y, p_z] GeV:
ℓ E= 2.2256 p_x= -0.0000 p_y= -2.2231 p_z= -0.0000
P E= 2.4129 p_x= 0.0000 p_y= 2.2231 p_z= 0.0000
ℓ' E= 1.9860 p_x= 0.0957 p_y= -1.9809 p_z= 0.0000
P' E= 2.4025 p_x= 0.0000 p_y= 2.2118 p_z= 0.0000
q_γ E= 0.2500 p_x= -0.0957 p_y= -0.2310 p_z= 0.0000
p_x [GeV]
p_y [GeV]
p_z [GeV]
P'
q_γ
ℓ
ℓ'
Transverse plane
Leading initial-ensemble/final-state components (N=3, retained=97.5%)
h_e=+1, h_p=+1, h_γ=-1
electron L, proton Ty
phase=-0.80, Re=1.10e+03, Im=-1.13e+03
h_e=+1, h_p=-1, h_γ=+1
electron L, proton Ty
phase=-1.76, Re=-2.67e+02, Im=-1.39e+03
h_e=+1, h_p=-1, h_γ=-1
electron L, proton Ty
phase=-0.10, Re=1.11e+03, Im=-1.06e+02
ensemble-weighted final-state |A|^2 fraction